



Digital transformation of the agri-food system

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The purpose of this paper is to examine the role of digital transformation in the agri-food sector. The study emphasizes digitalization as both an enabler of production efficiency and a radical innovator, redesigning business models and agricultural practices. The study explores the development of applications and products that connect consumers, supply chain actors, and producers, leading to customized food products. It highlights the notion of circular agri-food systems for feedback loops in the value chain, minimizing waste and integrating environmental and social values. Also, the paper explores the challenges in digital adoption, including technical barriers, privacy, and security concerns. To overcome these challenges, an interdisciplinary approach is proposed, merging technological, ecological, economic, and governance insights. Key needs identified for successful digital transformation include enhanced data processing, technological convergence, sustainability awareness, interoperability, and user adoption. The conclusion stresses the importance of invoking systemic thinking, user-friendly designs, and interdisciplinary collaboration in making sure that digital innovations enable a sustainable and resilient food production system.

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Introduction

A healthy, attractive, sustainable, and scalable food system is crucial for Green Transition and requires enabling Digital Transformation. Sustainable agri-food systems involve large-scale coordination of farming and value chain operations, resource use, and food chains,

and they require extensive digital devices and infrastructure technologies to enable and provide data and digital information chains [1]. Digitalization develops in two main pathways: 1) as an incremental enabler of production efficiency and sustainability metrics, policy monitoring, and so on, and 2) as an enabler of radical innovation and systemic changes, new redesigned processes, new business models, as a game-changer by changing farmer identity, farm work, farmer skills, ownership, privacy, and ethics [2].

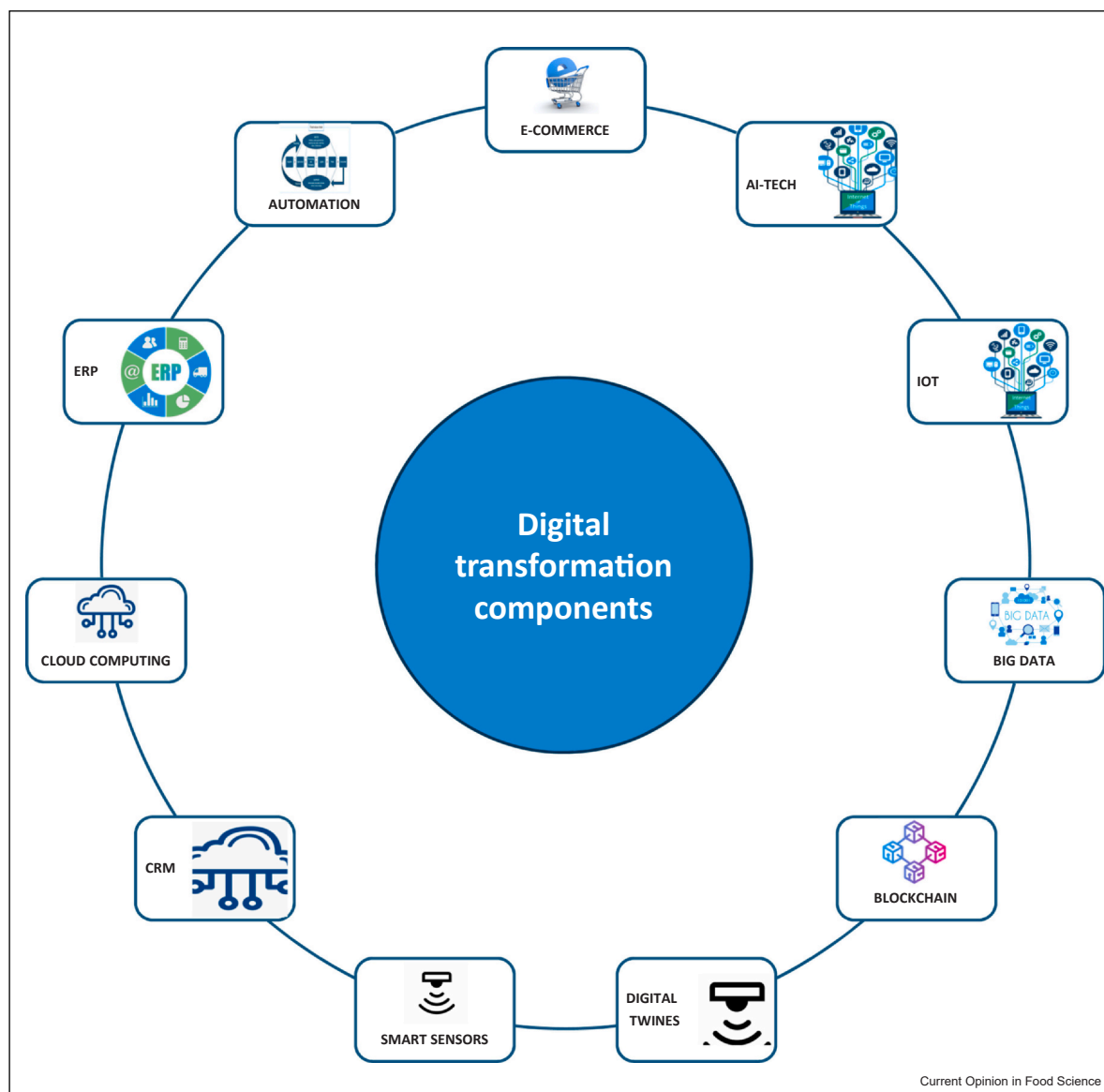
A key digital innovation is to enable/enhance the inter-linking (hyper-connecting) of consumers and producers in innovative ways by developing innovative applications, differentiating products, and reaching an advanced level of food product customization, feeding back information to the production. This creates an extended diversity of needs, requirements, and preferences. In this way, besides working with the farm-to-fork concept, engagement and demand-driven food production (from fork to farm) will be pursued at the same time [3]. Decision-making by various stakeholders in the food system will become more efficient as information flows will become transparent, providing only the information relevant to a specific decision [4].

Digital transformation

Digital transformation refers to a comprehensive change in business strategies, management models, and operations by incorporating digital technologies (DTs). It encompasses the digitalization of processes and the formation of extended supply chain relationships and making use of the internet in various aspects of the business. This transformation will lead to improved products and services, more efficient operations, cost reductions, and a competitive advantage in the market [5–7].

DTs are diverse and complex and often divided into four groups: efficiency technologies (e.g. Cloud Computing), connectivity technologies (e.g. Internet of Things [IoT]), trust disintermediation technologies (e.g. Blockchain), and automation technologies (e.g. Artificial Intelligence (AI) and Big Data) [8]. The process includes a ‘digitalization’ stage converting from analog to digital data, a ‘digitalization’ stage incorporating specific DTs into various business processes, and a final ‘digital transformation’ stage representing a comprehensive and widespread adoption of DTs, leading to new business models and a transformation of the logic and value-creation process. [Figure 1](#) represents some of the technologies applied in digital transformation.

Figure 1



Components of digital transformation.

Circular agri-food system

Agri-food systems encompass a wide range of activities, starting from the primary production of food and non-food agricultural products to their storage, post-harvest handling, transportation, processing, distribution, marketing, consumption, and disposal/reuse — see [Figure 2](#). Creating a circular agri-food system requires diverting end-of-life products from being discarded as waste and instead replacing primary input resources with secondary materials obtained through the reprocessing of these end-of-life products in closed-loop supply chains [9]. The creation of a circular system is closely linked to the concept of a circular economy, where a circular economy

is a regenerative system that minimizes resource waste, gas emissions, and energy leakage by slowing down, closing, and narrowing material and energy loops [10].

The circular economy can have a significant impact on the agri-food system. By adopting the principles of the circular economy, organizations can design an organizational architecture that not only considers economic value but also sustains the environmental and social value embedded in the product or service throughout its entire lifecycle [11]. By promoting the principles of reduction, reuse, and recycling, the circular economy aims to minimize waste and maximize resource efficiency in

Figure 2



Circular agri-food system.

all sectors, including agriculture and food production [12]. The circular economy can also lead to the development of more sustainable business models and technologies. Companies can adopt practices such as product-as-a-service and sharing platforms that extend the life of products and reduce the need for constant production of new goods. This can also reduce the environmental impact of DTs, such as e-waste and energy consumption [13]. The circular economy can help optimize resource use, reduce food waste, and improve overall sustainability. Farmers can adopt practices such as regenerative agriculture, which focuses on preserving and enhancing natural resources like soil health and

water systems. Food manufacturers can also find ways to reduce packaging waste and implement strategies for recycling and reusing food products [14].

Digital transformation is a requirement for developing a robust circular agri-food system by enabling more efficient and sustainable practices in the production, distribution, and consumption of food [15]. Blockchain has been widely explored in agri-food digital transformation. It can be used to create transparent and traceable supply chains, allowing consumers to track the journey of their food from farm to table. This can help build trust in the food system, reduce food fraud, and enable better

decision-making for sustainable sourcing [16]. Blockchain also plays a transformative role in smart contracts and trust-building by ensuring security, transparency, and automation in various industries. However, challenges like high implementation costs and scalability issues, adoption barriers and resistance from stakeholders, energy consumption and environmental concerns, and data accuracy and verification issues have limited their full potential.

Digital transformation can help identify inefficiencies in the food supply chain and facilitate the redistribution of surplus food to those in need. Technologies such as AI and machine learning can also predict and prevent food spoilage, reducing waste at all stages of the agri-food system [17]. AI enhances decision-making in agri-food systems by analyzing vast amounts of data from sensors, satellites, and market trends to guide farmers, processors, and supply chain managers. Moreover, AI-driven predictive analytics uses historical data and machine learning models to foresee and mitigate risks in agri-food systems. AI-powered automation can transform a system from static setting to dynamic and self-learning; it can reduce labor dependency, increase efficiency, and minimize natural resources and waste throughout the agri-food value chain. Digital platforms can facilitate the exchange of by-products and waste materials between different actors in the agri-food system, promoting circular economy practices such as recycling, composting, and energy recovery. This can help reduce the environmental impact of food production and create new revenue streams for farmers and businesses [18].

Digital transformation in the agri-food system

The agri-food sector is experiencing major transformations due to factors like the rapid growth of the global population, climate change, and economic globalization [5–7]. Contemporary terms such as Agriculture 4.0, Precision Agriculture, and Smart farming all refer to various forms of DT adoptions in this sector [5,19]. Therefore, to promote digital transformations in this field, it is crucial to understand the characteristics of the agri-food supply chain, like the perishable nature of its products, emphasis on customer satisfaction, and environmental impact [20]. The agri-food supply chain entails significant challenges impacting the entire chain, including sustainability issues like high energy and water consumption, land use and soil erosion, inadequate waste management, and greenhouse gas emissions [21]. Additionally, there are challenges related to food waste stemming from transportation, insufficient storage facilities, stringent export regulations, food safety and security such as food fraud, traceability discrepancies, and inadequate implementation of food security initiatives [22]. Table 1 represents different elements of the agri-food's value chain, such as

production, processing, distribution/consumption, and the disposal/reusing system with the impact of DTs and their application in agri-food systems.

Drivers of digital transformation in agri-food system

Digital transformation is a prerequisite and enabler for improving products and services, as well as the management of operations, making them more efficient. It also offers the potential for cost reduction and a competitive advantage in the market. The key drivers of digital transformation include technological advancements (for more examples, see Table 1, component of DT), increasing demand for efficiency and productivity, food safety and traceability, sustainability, resource management, changing market dynamics and competitiveness, connectivity, and infrastructure development (for more examples, see Table 1, impact of DT).

The agri-food industry is confronted with various market challenges, including the unpredictable nature of commodity prices, the ever-evolving consumer preferences, and the complex landscape of global trade regulations [20]. However, with the advent of digital transformation, businesses can now effectively navigate these challenges through the strategic use of data and insights [35]. DTs enable enterprises to swiftly adapt to market changes, make well-informed decisions, and offer personalized products and services to their customers [36]. This enables businesses within the agri-food industry to remain competitive in today's dynamic marketplace.

The ever-growing accessibility of internet connectivity, especially in rural regions, combined with the advancement in digital infrastructure, has opened up a world of possibilities for the digitization of agri-food systems. Farmers, agribusinesses, and various stakeholders can now tap into a wealth of valuable information, effortlessly communicate and collaborate, establish market connections, and harness the power of digital tools and platforms to enhance their operations.

The digital transformation of the agri-food sector is intertwined with geopolitical factors, necessitating adaptive strategies for resilience. Cross-border data flows and e-commerce, crucial to this transformation, are influenced by trade policies and regulations. Geopolitical events like trade wars and regional conflicts can disrupt supply chains, prompting the sector to leverage technologies such as IoT, blockchain, AI, and modeling for agility and resilience.

DTs, including marketplaces and blockchain, reduce reliance on traditional intermediaries (by combining smart contracts for automation and blockchain for trust-building, businesses and individuals can engage in

Table 1

The impact and application of DT in agri-food system.

Agri-food system	Impact of DT	Component of DT	Application of DT in agri-food	References
Production	Enhance efficiency, productivity, sustainability, and transparency	Smartphone applications, precision agriculture	Irrigation, fertilization, and harvesting of crops	[6,23,24]
	Remotely monitoring, tracking, and managing the performance of machineries	Global Navigation Satellite System (GNSS)	Crop's health and yield, livestock	[25]
Processing	Remotely monitoring, prediction, and decision-making processes related to the crop's health and livestock	IoT, sensors, drones, data analytics and machine learning algorithms	Soil moisture, and pest infestations	[26]
	Enhance productivity, reduce waste, enhance food safety, increase traceability and sustainable	IoT, sensors, actuators, and connected machineries, robots, and automation systems	Process optimization, anomalies detection	[6]
Distribution and consumption	Resilient and adaptive to changing market demands	Big Data, advanced analytics, and AI	Sorting, packing, and quality control	[7,27]
	Reducing labor costs, enhancing speed and accuracy, and minimizing the risk of accidents	Robots and automation systems, AI, sensors	Sorting, packing, and quality control	[28]
	Improve efficiency, transparency, and sustainability	IoT, AI, blockchain, data analytics, and mobile applications	Supply chain management	[27]
	Real-time monitoring, tracking food products during transport, ensuring food safety and quality	Blockchain, IA, sensors	Inventory management and transportation routing	[7]
	Prevent foodborne illnesses, reduce the risk of recalls, and build consumer trust in the quality and safety of food products	IoT, sensors	Supply chain management	[29]
Food disposal/reusing	Predict demand, optimize logistics, identifying inefficiencies, and automating tasks, facilitate personalized marketing and direct-to-consumer sales	Online marketplaces and delivery platforms, AI algorithms	Supply chain management, Personalized recommendations, and targeted advertising	[30]
	Optimize waste management, enhancing sustainability and efficiency	IoT, sensors	Waste management	[6]
	Optimize processes and predict future waste, aiding efficient resource allocation	Advanced analytics and machine learning	Waste generation, disposal/reuse flows	[31]
	Facilitating surplus food redistribution and reducing waste	Digital platforms	Stakeholders' connection	[32]
	Ensures traceability, transparency, and tracking food origins	Blockchain, mobile apps, and web tools	Waste management	[33]
	Enhancing efficiency and recycled material quality	Robotic systems	Waste facilities automate sorting, recycling, and composting	[34]

secure, transparent, and efficient transactions without relying on intermediaries), but geopolitical protectionism can impose data localization laws, impacting cloud-based and data-sharing solutions. Conversely, geopolitical collaborations offer opportunities for knowledge exchange and technology transfer, fostering sectoral innovation. The sector's growth is also tied to e-commerce, the online trading model that aligns it with global economic and political trends.

In the digitally transforming agri-food sector, securing data is vital, especially with rising geopolitical tensions heightening cyber threats. Robust cybersecurity is crucial to protect sensitive data and critical infrastructure against these threats [37]. Blockchain technology is particularly beneficial, enhancing traceability and reducing fraud in cross-border e-commerce and food supply chains [38]. Keeping hardware and software updated is essential for cybersecurity. With increased awareness and stricter data protection regulations, including mandatory breach notifications and higher fines, the sector is expected to align its digital transformation with these cybersecurity demands. Companies need to navigate these regulations while ensuring data security and privacy compliance. However, failing to adopt cybersecurity practices to the agri-food sector's unique structure and complexities could lead to vulnerabilities, risking food supply chain disruptions. As reliance on DT grows in agriculture, cybersecurity risks underscore the need for significant investment in cybersecurity research and development to safeguard the food and feed supply.

Challenges of digital transformation in agri-food systems

One of the primary barriers to successful digital transformation is the concern over data privacy and security. Both public and private organizations regularly encounter difficulties in guaranteeing the security and privacy of data shared during digital transformation initiatives [39]. This can result in a lack of trust among stakeholders and impede collaboration. Establishing trust among stakeholders is essential for the effective implementation of digital transformation [40]. Issues such as a lack of transparency in data exchange processes and governance can create obstacles to collaboration and impede progress. Additionally, power dynamics within supply chains can pose a significant barrier to successful digital transformation efforts [41]. Companies may encounter resistance from suppliers or partners who are reluctant to embrace new technologies or processes.

The digital divide is yet another obstacle to transformation, referring to the gap between those with access to DTs and those without [42]. This discrepancy can hinder digital transformation, as organizations may struggle to engage with stakeholders who lack digital

literacy or essential technological infrastructure. Additionally, inadequate technology infrastructure can impede digital transformation efforts, especially in regions with limited connectivity or outdated systems.

Organizations may face challenges implementing new technologies or processes without the necessary infrastructure in place [43]. Integrating various systems and technologies can also be complex, particularly when systems are not compatible. This lack of interoperability can create barriers to data exchange and collaboration between organizations. Inconsistent data formats and standards can further impede the exchange of information, resulting in inefficiencies and errors. Achieving data interoperability is crucial for successful digital transformation initiatives [44]. Organizations operating across borders must navigate a complex regulatory environment, characterized by varying data protection and privacy laws in different countries [45]. A deeper engagement with governance and policy aspects, including interoperability, regulations, and compliance, is obvious for ensuring efficiency, legal certainty, and sustainable growth.

The digitalization of the agri-food system faces key academic challenges. These include designing IT systems with a clear purpose, specifying accurate data requirements, utilizing big data analytics, developing digital twins, and ensuring data interoperability and standardization. Moreover, there is a lack of reference and ground truth data for validating sensor data and enhancing AI capabilities in agriculture. The nature of diversified and circular agri-food systems further complicates these challenges, requiring innovative and adaptable digital solutions.

Digitalization requires an interdisciplinary approach, combining expertise from data science, computer science, system engineering (including AI, Machine Learning, and Big Data), and communication technologies. Additionally, it demands insights from social sciences, economics, business, and organizational studies to understand its broader impacts. Knowledge in production and operations management, along with biological science and domain-specific expertise, is crucial. Importantly, digitalization must be evaluated at a system level, not just at the device level, to ensure holistic and sustainable integration across various domains.

Digital transformation of agri-food systems requires a systemic approach to digitalization that looks beyond individual technologies to explore interconnected technological systems, such as IoT and sensor networks, and data ecosystems, as well as how they interact with agricultural, ecological, economic, and regulatory systems. This requires integration across user-level digitalization, farm-level digitalization, agri-food chain digitalization, and policy-level digitalization. This multilevel integration must ensure a comprehensive digital transformation,

addressing the complexities of the agri-food sector for more efficient, sustainable, and resilient food systems.

Given the complex challenges within the agri-food value chain, particularly in areas such as real-time replanning and optimization, it is evident that DT still faces obstacles in efficiently processing vast amounts of data, adapting to dynamic conditions, and providing actionable solutions promptly. Interoperability and standardization of data formats also pose significant hurdles, as data from various sources often differ in structure and frequency, making cohesive integration a daunting task.

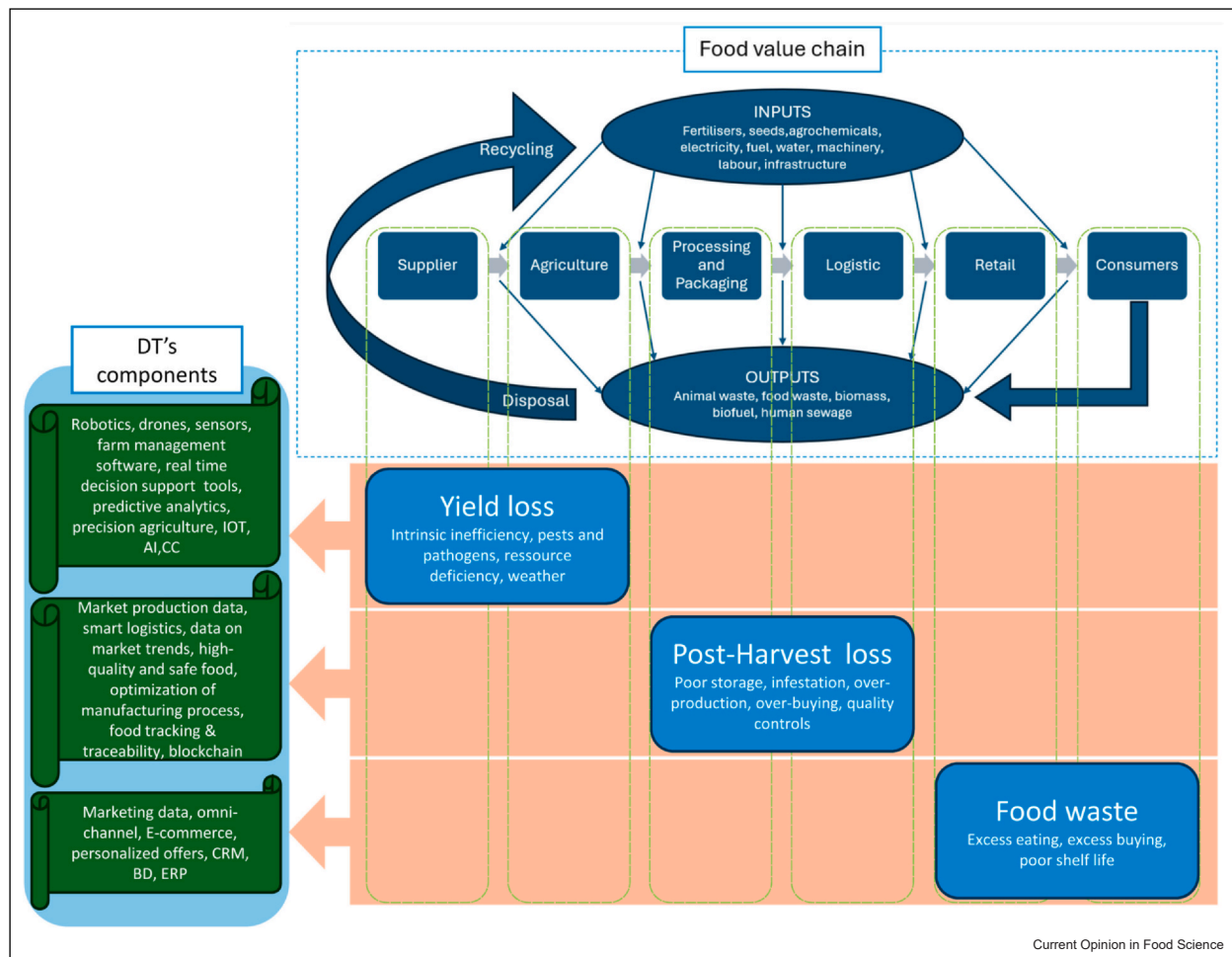
Moreover, optimization algorithms, particularly those leveraging AI and machine learning, can be computationally intensive, adding another layer of complexity to the digitalization of agriculture. Data transfer limitations also present a critical issue, especially in large-scale systems, amplifying the complexity and flow of data within the agri-food system.

Coordinating multiple interconnected entities in real-time, such as vehicles, machinery, and personnel, further compounds the challenges of synchronization and decision-making. Additionally, developing user-friendly interfaces for decision-making tools remains a crucial concern, especially in human-in-the-loop systems where operators rely on clear, understandable data outputs for quick decision-making.

The traceability of products within the agri-food system, aided by blockchain technology, is another critical issue that has not been fully realized due to infrastructure limitations and cost constraints. Furthermore, issues surrounding data ownership, regulations, and policies in data spaces present ongoing challenges that must be addressed in order to fully unlock the potential of DT in the agricultural sector.

Based on [Figure 3](#), it can be concluded that the value chain of a circular agri-food system is intricate, involving

Figure 3



Overview of digital transformation in agri-food system.

numerous stakeholders with varying levels of interest in digitalization. Implementing digital transformation components is the key to addressing the complexities and challenges within this system.

Synthesis and conclusion

This review has made a key contribution by identifying the challenges and opportunities of digital transitions throughout the entire value chain in the agri-food system, spanning from production to disposal. In addition, it examines the required components of digital transformation specifically applied in the agri-food industry. The analysis presented in this review effectively identifies key factors driving or hindering the digital transformation process within the agri-food system.

This review lays the baseline for the future debate by recommending a systems approach towards digitalization, which refers to the comprehensive integration of DTs into all aspects of an organization or system, considering not only individual technologies but also their interconnections, processes, and the overall ecosystem. This review shows that, instead of using isolated tools, the system approach emphasizes the importance of the interconnectedness of digital tools, data, infrastructure, and workflows. This review concludes that by adopting a systems thinking perspective, digitalization benefits the entire organization by improving workflows, reducing costs, and enhancing customer experiences. This approach considers long-term sustainability, scalability, and collaboration with partners and stakeholders to create a networked digital ecosystem. This approach allows organizations to remain agile and adaptable, responding quickly to market changes and technological advancements.

The authors assess, based on this review, that the digital transformation of the agri-food system still needs several improvements and requirements. One is enhanced data processing/data sharing, which creates the foundation for better decision-making. Secondly, efficient processing and sharing of data enables stakeholders to make informed choices based on real-time information and trends, which are formed by four needs. The first need is the convergence of technologies, where various technologies are integrated, such as IoT, AI, and blockchain, to work seamlessly together and lead to more efficient and effective solutions in agricultural practices. The second is digitalization for sustainable food production awareness, where digital tools are used to quantify sustainability indicators, raising awareness about the sustainability impact of food production, including tracking carbon footprints, water usage, and other sustainability metrics. The third is improved interoperability, standards, and semantics to ensure that different digital systems and tools can communicate and work together

effectively as a crucial prerequisite for a unified digital approach. The fourth is enhanced user adoption focusing on how users, especially farmers and food producers, adopt and interact with technology and involve user-friendly designs, training, and support systems.

Addressing these needs requires an interdisciplinary approach that goes beyond technology itself. It involves understanding human-to-technology interactions, changing mindsets about how digitalization integrates with production systems, and focusing on implementing digital systems from a total system perspective. This comprehensive approach ensures that digital transformation not only enhances efficiency but also supports sustainable and resilient food production systems.

Data Availability

No data were used for the research described in the article.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationship that could have appeared to influence the work reported in this paper.

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References and recommended reading

Papers of particular interest, published within the period of review, have been highlighted as:

- of special interest
- of outstanding interest

1. El Bilali H, Strassner C, Ben Hassen T: **Sustainable agri-food systems: environment, economy, society, and policy.** *Sustainability* 2021, **17**:1667.
2. Balogun AL, Marks D, Sharma R, Shekhar H, Balmes C, Maheng D, Arshad A, Salehi P: **Assessing the potentials of digitalization as a tool for climate change adaptation and sustainable development in urban centres.** *Sustain Cities Soc* 2020, **53**:101888.
3. Reinhardt T: **The farm to fork strategy and the digital transformation of the agrifood sector — an assessment from the perspective of innovation systems.** *Appl Econ Perspect Policy* 2023, **45**:819-838.
4. Wang R, Bush-Evans R, Arden-Close E, Bolat E, McAlaney J, Hodge S, Thomas S, Phalp K: **Transparency in persuasive technology, immersive technology, and online marketing: facilitating users' informed decision making and practical implications.** *Comput Hum Behav* 2023, **139**:107545.
5. Hassoun A, Marvin HJP, Bouzembrak Y, Barba FJ, Castagnini JM, •• Pallarés N, Rabail R, Aadil RM, Bangar SP, Bhat R, et al.: **Digital transformation in the agri-food industry: recent applications and the role of the COVID-19 pandemic.** *Front Sustain Food Syst* 2023, **7**:127813.

Publication explores the integration of DTs in the agri-food industry, emphasizing the impact of COVID-19 in driving adoption. The study discusses the benefits, challenges, and practical applications of these technologies, highlighting the role of Industry 4.0 innovations. It also presents recommendations for achieving inclusive digital transformation in the food industry.

6. Sridhar A, Ponnuchamy M, Kumar PS, Kapoor A, Nguyen Vo DV, Rangasamy G: **Digitalization of the agro-food sector for achieving sustainable development goals: a review.** *Sustain Food Technol* 2023, **1**:783-802.
Publication highlights the significance of food security and agricultural sustainability in the face of rising food demand, inequality, climate change, and pandemics. The authors suggest that DTs can solve challenges in the agro-food sector, enhance production and supply chain, and achieve sustainable development goals. They propose integrating AI, precision farming, and big data analytics to boost yield and sustainability. Digitalization in agriculture can also cut costs and waste, while improvements in food safety and electronic commerce are anticipated, with potential advancements in blockchain technology for improved traceability and personalized nutrition.
7. Balyan S, Jangir H, Tripathi SN, Tripathi A, Jhang T, Pandey P: **Seeding a sustainable future: navigating the digital horizon of smart agriculture.** *Sustainability* 2024, **16**:475.
Publication highlights the importance of agriculture for human survival but acknowledges that it is facing challenges due to climate change. The agriculture industry is using various measures and technologies to adapt, including plant breeding, robotics, and big data. The digitalization wave offers opportunities for smart farming but also presents challenges such as high implementation costs and data security concerns. The shift to digital farming offers solutions for environmental issues and economic growth, with future advancements potentially including data encryption and economic policies.
8. Santos FJ, Guzmán C, Ahumada P: **Assessing the digital transformation in agri-food cooperatives and its determinants.** *J Rural Stud* 2024, **105**:103168.
9. Miranda BV, Monteiro GFA, Rodrigues VP: **Circular agri-food systems: a governance perspective for the analysis of sustainable agri-food value chains.** *Technol Forecast Soc Change* 2021, **170**:120878.
10. Geissdoerfer M, Pieroni MPP, Pigosso DCA, Soufani K: **Circular business models: a review.** *J Clean Prod* 2020, **277**:123741.
11. Farace B, Tarabella A: **Exploring the role of digitalization as a driver for the adoption of circular economy principles in agrifood SMEs — an interpretive case study.** *Br Food J* 2024, **126**:409-427, <https://doi.org/10.1108/BFJ-12-2022-1103>.
Publication explores how digitalization can help integrate circular economy principles in the Italian agrifood sector, using a case study of a small and medium enterprise. The findings show that digitalization optimizes resource use, minimizes waste, and interacts with sustainability and resilience concepts. The study also highlights the importance of digital transition in the agrifood industry, which faces resistance due to the size of firms and managerial conservatism tradition.
12. Renaldi V: **Sustainable energy transition: circular economy and sustainable financing for environmental, social and governance (ESG) practices.** *Soc Environ Account J* 2024, **44**:266-269, <https://doi.org/10.1080/0969160X.2024.2401212>
13. Channi HK, Kumar R: **Chapter 14 — Digital technologies for fostering sustainability in Industry 4.0.** In *Evolution and Trends of Sustainable Approaches*. Edited by Rossit DA, Hussain CM. Elsevier; 2024:227-251.
14. Cahyadi ER, Hidayati N, Zahra N, Arif C: **Integrating circular economy principles into agri-food supply chain management: a systematic literature review.** *Sustainability* 2024, **16**:7165.
15. Sica D, Esposito B, Malandrino O, Supino S: **The role of digital technologies for the LCA empowerment towards circular economy goals: a scenario analysis for the agri-food system.** *Int J Life Cycle Assess* 2024, **29**:1486-1509.
16. Oriekhoe OI, Ilugbusi BS, Adisa O: **Ensuring global food safety: integrating blockchain technology into food supply chains.** *Eng Sci Technol J* 2024, **5**:811-820.
17. Suali AS, Srai JS, Tsolakis N: **The role of digital platforms in e-commerce food supply chain resilience under exogenous disruptions.** *Supply Chain Manag* 2024, **29**:573-601.
18. Viscardi S, Colicchia C: **A classification of food products to enhance circular economy and reduce waste: a systematic literature review.** *Resour Conserv Recycl Adv* 2024, **23**:200229.
19. Dayioğlu MA, Türker U: **Digital transformation for sustainable future-agriculture 4.0: a review.** *Tarim Bilim Derg* 2021, **27**:373-399.
The article discusses the challenges faced by agriculture, such as global warming and climate change. It emphasizes the need for digital solutions and outlines the transition from traditional to digital agriculture. Various technologies and applications are explained, and the authors conclude that innovations can contribute to a sustainable future.
20. Yadav VS, Singh AR, Gunasekaran A, Raut RD, Narkhede BE: **A systematic literature review of the agro-food supply chain: challenges, network design, and performance measurement perspectives.** *Sustain Prod Consum* 2022, **29**:685-704.
Publication discusses the challenges faced by the agriculture food supply chain (AFSC), including food safety, sustainability, government regulation, and traceability issues. It also reviews research on designing the agro-food supply chain network and investigates performance measurement using 108 articles. The article highlights the potential of technology adoption and integration, as well as the digital transformation of AFSC. It emphasizes other research trends and identifies possible future challenges.
21. Joshi S, Singh RK, Sharma M: **Sustainable agri-food supply chain practices: few empirical evidences from a developing economy.** *Glob Bus Rev* 2023, **24**:451-474.
22. Islam N, Nazir W, Khalid N: **Conceptual study of problems and challenges associated with the food supply chain in developing countries.** *Arab Gulf J Sci Res* 2021, **39**:100-117.
23. Fuentes-Peñailillo F, Gutter K, Vega R, Silva GC: **Transformative technologies in digital agriculture: leveraging Internet of Things, remote sensing, and artificial intelligence for smart crop management.** *J Sens Actuator Netw* 2024, **13**:39.
24. Papadopoulos G, Arduini S, Uyar H, Psiroukis V, Kasimati A, Fountas S: **Economic and environmental benefits of digital agricultural technologies in crop production: a review.** *Smart Agric Technol* 2024, **8**:100441.
25. McCampbell M: **Give to AgEcon Search Agricultural digitalization and automation in low-and middle-income countries: Evidence from ten case studies;** 2022.
26. Vinod Chandra SS, Anand Hareendran G, Albaaji GF: **Precision farming for sustainability: an agricultural intelligence model.** *Comput Electron Agric* 2024, **226**:109386.
27. Konfo TRC, Djouhou FMC, Hounhouigan MH, Dahouenon-Ahoussi E, Avlessi F, Sohounhloue CKD: **Recent advances in the use of digital technologies in agri-food processing: a short review.** *Appl Food Res* 2023, **3**:100329.
Publication discusses the recent advances in using DTs in agri-food processing to meet the increasing demand for food while addressing sustainability concerns. These technologies offer benefits such as increased efficiency and reduced waste and enable real-time monitoring of critical parameters to ensure food quality and safety. The review also covers the challenges and opportunities for adopting DTs in the industry and highlights its potential for revolutionary transformation.
28. Yap CK, Al-Mutairi KA: **A conceptual model relationship between Industry 4.0 — food-agriculture nexus and agroecosystem: a literature review and knowledge gaps.** *Foods* 2024, **13**:150.
29. Chhetri KB: **Applications of artificial intelligence and machine learning in food quality control and safety assessment.** *Food Eng Rev* 2024, **16**:1-21.
30. Haleem A, Javaid M, Asim Qadri M, Pratap Singh R, Suman R: **Artificial intelligence (AI) applications for marketing: a literature-based study.** *Int J Intell Netw* 2022, **3**:119-132.
31. da Costa TP, Gillespie J, Cama-Moncunill X, Ward S, Condell J, Ramanathan R, Murphy F: **A systematic review of real-time monitoring technologies and its potential application to reduce food loss and waste: key elements of food supply chains and IoT technologies.** *Sustainability* 2023, **15**:614.
32. Schroder A, Prockl G: **Ioanna Constantiou: 53rd Hawaii International Conference on System Sciences (HICCS), January 5–8, 2021. University of Hawai'i at Manoa; 2021.**

33. Pakseresht A, Yavari A, Kaliji SA, Hakelius K: **The intersection of blockchain technology and circular economy in the agri-food sector**¹. *Sustain Prod Consum* 2023, **35**:260-274.
34. Salem KS, Clayson K, Salas M, Haque N, Rao R, Agate S, Singh A, Levis JW, Mittal A, Yarbrough JM, et al.: **A critical review of existing and emerging technologies and systems to optimize solid waste management for feedstocks and energy conversion**. *Matter* 2023, **6**:3348-3377.
35. Choudhary L, Swami S: **Harnessing the power of big data: revolutionizing agriculture**. *Curr J Appl Sci Technol* 2023, **42**:40-49.
 Publication highlights the utilization of big data in the agriculture industry to improve productivity and addresses challenges such as global food crises. Smart farming leverages big data for decision-making, efficiency, and resource optimization. However, safeguarding sensitive data presents ethical and legal challenges. The digital revolution in agriculture is promising and irreversible.
36. Alojail M, Khan SB: **Impact of digital transformation toward sustainable development**. *Sustainability* 2023, **15**:14697.
37. Nobles C, Burrell DN, Waller T, Cusak A: **Food sustainability, cyber-biosecurity, emerging technologies, and cybersecurity risks in the agriculture and food industries**. *Int J Environ Sustain Green Technol* 2022, **13**:1-17.
38. Antonucci F, Figorilli S, Costa C, Pallottino F, Raso L, Menesatti P: **A review on blockchain applications in the agri-food sector**. *J Sci Food Agric* 2019, **99**:6129-6138.
39. Stewart H: **Digital transformation security challenges**. *J Comput Inf Syst* 2022, **63**:919-936.
40. Barrane FZ, Ndubisi NO, Kamble S, Karuranga GE, Poulin D: **Building trust in multi-stakeholder collaborations for new product development in the digital transformation era**. *Benchmark Int J* 2021, **28**:205-228.
41. dos Santos Silva J, Matos de Oliveira A, Verissimo de Oliveira J, Bouzon M: **Barriers to digital transformation in fruit and vegetable supply chains: a multicriteria analysis using ISM and MICMAC**. *OPSEARCH* 2025, **62**:460-482, <https://doi.org/10.1007/s12597-024-00809-6>
42. Samuel-Okon AD, Abejide OO: **Bridging the digital divide: exploring the role of artificial intelligence and automation in enhancing connectivity in developing nations**. *J Eng Res Rep* 2024.
43. Hendrawan1 * SA, Chatra A, Iman N, Hidayatullah S, Suprayitno D: **Digital transformation in MSMEs: challenges and opportunities in technology management**. *J Inf Dan Teknol* 2024, **6**:141-149.
44. Ahmed O, Yazdeen AA, Mohammed T, Sami G: **Consideration of Cloud-Web-Concepts for Standardization and Interoperability A Comprehensive Review for Sustainable Enterprise Systems, AI, and IoT Integration**. *JITI*; 2024.
45. Akanfe O, Lawong D, Rao HR: **Blockchain technology and privacy regulation: reviewing frictions and synthesizing opportunities**. *Int J Inf Manag* 2024, **76**:102753.